

Harshit Mittal

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Education

University of Leeds

September 2025 - August 2026

Master of Science in Advanced Computer Science (First Class)

Leeds, United Kingdom

- **Relevant Coursework:** Algorithms (C++), Machine Learning (Python), Deep Learning (Python), Data Mining and Text Analysis (Python), Data Science (Python).
- **Thesis:** The design and implementation of SAFViT, a novel hierarchical encoder-decoder architecture integrating Spatial Adaptive Fusion over the sliding window backbone, outperforming models like CellViT across minority dead cells on the PanNuke dataset.

Maharaja Agrasen Institute of Technology

August 2021 - May 2025

Bachelors of Technology in Computer Science and Engineering (First Class)

Delhi, India

- **Relevant Coursework:** Data Structures (C++), Algorithms (C++), Machine Learning (Python), Data Science (Python), Natural Language Processing (Python), Software Development, Advanced Mathematics, Advanced Physics.

Experience

USIC&T

Jan 2025 – May 2025

AI Research Intern

Delhi, India

- Pioneered novel medical AI activation and class-balancing techniques, slashing information loss to 7% and boosting accuracy by up to 10% across peer-reviewed frameworks (Springer/IEEE).
- Architected containerized ML pipelines using Docker and Git, standardizing cross-team reproducibility and streamlining version-controlled feature deployments.

CFEES, DRDO

Aug 2023 – Oct 2023

Project Trainee

Delhi, India

- Engineered and deployed a peer-reviewed 3D facial recognition pipeline that upgraded CCTV footage to point-clouds, directly improving the department's active production security posture.
- Optimized cross-platform Flutter application performance to boost end-user satisfaction, establishing rigid code standards through meticulous Git history and documentation tracking.

Projects

AdaptAttend – Adaptive Attendance System | Flutter, Dart, TFLite, ML Kit, Drift, Next.js | [🔗](#)

- Built a fully offline Flutter attendance app supporting 3 methods – 6-digit TOTP, GPS geo-fence, and manager-side face scan – with zero backend dependency; all data stored on-device via Drift (SQLite).
- Integrated MobileFaceNet TFLite model (128-d embeddings, cosine similarity) with Google ML Kit face detection for on-device 1-to-N face recognition, replacing manual PIN entry and eliminating remote attendance fraud.
- Engineered a geo-fence security layer using Haversine distance to block attendance marking outside a configurable office radius, and restricted face scan to the manager's device to prevent spoofing.

ShopHandBook | Flutter, Dart, SQLite, pdf package, fl_chart, Provider | [🔗](#)

- Engineered a fully offline Android shop management app with 5 integrated modules – document scanner, PDF billing, inventory manager, customer tracker, and analytics – eliminating the need for internet connectivity or third-party cloud services.
- Implemented on-device multi-page PDF generation using the pdf package, enabling users to scan and merge multiple camera captures into a single invoice or document with zero server dependency.
- Built a local SQLite database layer via a singleton DatabaseService with full CRUD, relational bill-item joins, and dashboard aggregation queries, supporting complete shop data management entirely on-device.

Technical Skills

Languages: Python, Dart, C, C++, R, JavaScript, HTML, CSS.

Technologies: Flutter, PyTorch, TensorFlow, Azure AI, TFLite, Docker, Git, SQLite, Kubernetes.

Concepts: Deep Learning, Computer Vision, Face Recognition, Transformers, On-device ML, Agile Methodology, CI/CD, Cloud Computing.